

Questions: Solving first-order linear ODEs using integrating factors

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Summary

A selection of questions for the study guide on solving first-order linear ODEs using integrating factors.

Before attempting these questions, it is highly recommended that you read [Guide: Introduction to solving first-order linear ODEs using integrating factors](#).

Q1

Rearrange each first-order linear ODE below to be in the form, $\frac{dy}{dx} + P(x)y = Q(x)$ with respect to the variables in each equation.

1.1. $-\frac{dy}{dx} = y + 8.$

1.2. $\frac{1}{2x} = \frac{\frac{dy}{dx} + 5x}{y}.$

1.3. $\frac{dx}{dt} - \cos(t) = -x \cos(t).$

1.4. $x \frac{dy}{dx} - x^2 = y.$

Q2

Find the integrating factor of every question in question 2.

Q3

Solve each equation in question 2 for the general solution.

Q4

4.1. Find the exact solution to question 1.1. given the initial condition $y(0) = 0.$

4.2. Find the exact solution to question 1.2. given the initial condition $y(1) = 6.$

4.3. Find the exact solution to question 1.3. given the initial condition $x(0) = \pi$.

4.4. Find the exact solution to question 1.4. given the initial condition $y(2) = 3$.

Q5

5.1 Find the general solution to $\frac{dy}{dx} - \frac{5}{x}y = 3x + 2$.

5.2 Find the general solution to $\frac{dy}{dx} - \frac{-4y}{4x+1} = 5x^4$.

5.3 Find the exact solution to $\frac{dy}{dx} - \cos(x) = -4y \cot(x)$ when $y(\frac{\pi}{2}) = 1$.

5.4 Find the exact solution to $e^{4x} \frac{dy}{dx} + 4e^{4x}y = 4x \sin(x)$ when $y(0) = 4$.

[After attempting the questions above, please click this link to find the answers.](#)

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v1.0: initial version created 04/06 by mak.

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